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Interfacial Structures in Ionic Liquid Based Ternary Electrolytes for Lithium-Metal Batteries: a Molecular Dynamics Study

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Abstract

Lithium-metal batteries (LMB) are promising candidates to fulfill the future performance requirements for energy storage applications. However, the tendency to form metallic dendrites and the undesirable side-reactions between electrolyte and the Li electrode lead to poor performance and safety issues in these batteries. Thereby, understanding the interfacial properties and the Li-metal surface/electrolyte interactions are crucial to resolve the remaining obstacles and make these devices feasible. Here we report a computational study on the interface effects in ternary polymer electrolytes composed by poly(ethylene oxide) (PEO), lithium salts and different ionic liquids (ILs) confined between two Li-metal slabs. Atomistic simulations are used to characterize the local environment of the Li^+ ions and the transport properties in the bulk and at the interface regions. Aggregation of ions at the metal surface are seen in all investigated systems, the structure and composition are directly correlated to the IL components. The strong interactions between the electrolyte species and the Li-metal atoms result in the structuring of

the electrolyte at the interface region, in which comparatively small and flat ions result in a well defined region with extensive Li^+ populations and high self-diffusion coefficients. In contrast, large ions such as $[\text{P222mom}]^+$ increase the PEO density in the bulk due to large steric effects at the interface. Therefore, the choice of specific ILs in ternary polymer electrolytes can tune the structure-dynamic properties at the Li-metal surface/electrolyte interface, controlling the SEI formation at the electrode surface, and thereby improve battery performance.

1 Introduction

The need for reducing the dependence on fossil fuels and associated environmental issues has led to excessive demand for high-performance batteries, e.g., superior to conventional Li-ion batteries (LIBs). It has been estimated that batteries with energy densities larger than 500 Wh kg^{-1} will be required to meet the future battery demands.^{1,2} In this context, lithium-metal batteries (LMBs), i.e., utilizing Li-metal as the negative electrode, are one of the most promising candidates. Li-metal can deliver a theoretical capacity of 3860 mA h g^{-1} ,³

which is more than ten times higher than that of graphite electrode in the commercial Li-ion battery (LIB), and an unprecedentedly low electrochemical potential (-3.04 V vs the standard hydrogen electrode). Thereby, LMBs possess high energy density.^{3,4}

Despite the aforementioned advantages, the high reactivity of lithium metal with the electrolyte, the tendency to form metallic dendrites due to uneven Li deposition during charging, and the inability of forming a robust solid electrolyte interphase (SEI) layer constitute the major challenges for the successful application of LMBs.^{5,6} The continuous dissolution/formation of the SEI during electrochemical cycling results in consumption of the electrolyte and inhomogeneous Li^+ deposition at the metal surface and, consequently, lithium dendrites growth.⁷ These drawbacks lead to poor battery performance and safety issues due to internal short circuits in the cell.

The development of new electrolytes is the key to stabilize and make LMBs feasible.⁶ In this context, ionic liquids (ILs) based electrolytes form a stable and passive electrolyte interface that decreases the dendrites growth in the battery.^{8–10} The use of IL electrolytes also increase the battery's Coulombic efficiency¹¹ and enables a better control of the side reactions in, for example, Li–S batteries.¹² Beyond ILs, solid polymer electrolytes (SPEs)^{13–17} also show great applicability for the LMBs. Their elastic modulus serve as a physical barrier that suppress dendrite growth. However, SPEs often present low ionic conductivity and limited electrochemical windows.

In the last years, several studies have focused on the development of hybrid SPEs,^{18–23} e.g., mixtures of polymers and ILs,^{24–28} as one alternative to improve the performance of SPEs. The advantage of these materials is the possible combination of the physicochemical properties of the ILs and polymers: while the ILs render decent ionic conductivities, useful electrochemical stability window and stable interfacial interactions, the polymer chains provide mechanical stability together with assisting the Li^+ transport mechanism by the

weakening of the ionic interactions and through segmental dynamics of the polymer chains.²⁵

Mixtures of poly(ethylene oxide) (PEO), Li salts and ionic liquids, are by far the most studied hybrid IL/SPEs systems.^{13,29–31} The addition of ionic liquids into PEO-based electrolytes can render ionic conductivities ten times larger than in the IL free systems,³² anodic limits up to 5 V vs. Li/Li^+ ³³ and long-term stability with Li-metal anode.²⁷

Costa *et al.* have conducted a series of molecular dynamics (MD) simulations for several PEO and ILs blends.^{22,34–36} In these ternary polymer electrolyte systems, i.e., (IL/PEO/LiSalt), the Li^+ local environment is mainly dominated by the interactions between the Li^+ and the O atoms from the PEO chains. The presence of the IL in the systems leads to a plasticizing effect on the polymer due to the interactions between the ions and PEO, leading to an increase of the chain dynamics and a better electrolyte performance.²²

Although IL/PEO/Li-salt systems have been widely investigated in the literature, by far most studies focus only on the bulk electrolyte properties, and rather similar ILs.^{22,37,38} However, it is well known that small changes in the IL ionic structure^{32,33,38} has a direct impact on the electrolyte performances. Besides, the interactions between the electrode and the electrolyte atoms lead to different electrolyte properties at the bulk and in the interface regions,³⁹ which can affect the overall battery performance.

Thus, it is necessary to understand the effect of the different IL ionic structures in the ternary polymer electrolyte performance, along with the nanostructure of the electrolyte at the electrode surface, to reach a microscopic understanding about these interfacial phenomena.⁴⁰ In the case of the LMBs, such understanding will serve as a first step to render a stable battery system. Therefore, in this present work, we have carried out classical MD simulations to characterize and evaluate the interfacial and bulk properties between ternary polymer electrolytes and the Li-metal surface. While a significant amount of studies have focused on IL and electrolytes at interfaces,^{39,41–44} to the best of our knowledge,

this is the first MD study concerning the interfaces effects between a IL/PEO/Li-salt electrolyte and a Li-metal surface.

The MD simulations were performed for three electrolytes based on the combination of either the cations 1-ethyl-3-methylimidazolium ($[\text{C}_2\text{mim}]^+$) or (methyloxymethyl)triethylphosphonium ($[\text{P222mom}]^+$), the anions 1,2,3-triazolide $[\text{123Triaz}]^-$ or bis(trifluorsulphonyl)imide $[\text{TFSI}]^-$, together with Li^+ ions and PEO. The systems compositions are listed in Table 1.

The choice of ILs is not only based on their applicability as IL electrolytes,^{45–50} but also due to their structural differences and the previous reported ability of them to form a stable electrolyte interface at the Li-metal surface. By using these systems, it is possible to rationalize the influence of the size and shapes of the ions on the interfacial properties, as well as the electrolyte performance, through the interchange of $[\text{P222mom}]^+$ for $[\text{C}_2\text{mim}]^+$, or $[\text{123Triaz}]^-$ for $[\text{TFSI}]^-$, in the electrolyte composition

2 Theoretical approach and computational details

2.1 Force field and models

The MD simulations employed the General Amber Force Field (GAFF),⁵¹ as implemented in the Groningen Machine for Chemical Simulations (GROMACS) package,^{52,53} version 2018.3, which has been widely employed to describe the thermodynamics and structural properties of several ionic liquids and polymer electrolytes at both the bulk and interface regions.^{54–56} Based on the GAFF potential, the total energy, U_{tot} , at zero temperature is given by the equation below,⁵¹

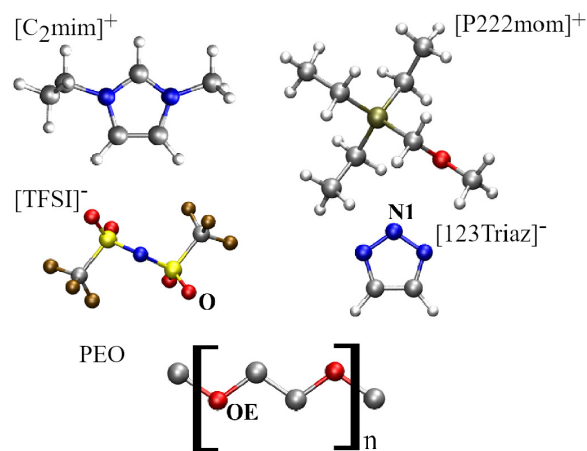


Figure 1: Ions and polymer models used in the simulations. 1-ethyl-3-methylimidazolium ($[\text{C}_2\text{mim}]^+$), (methyloxymethyl)triethylphosphonium ($[\text{P222mom}]^+$), 1,2,3-triazolide ($[\text{123Triaz}]^-$), bis(trifluorsulphonyl)imide ($[\text{TFSI}]^-$) and poly(ethylene oxide) (PEO) chains. C, H, O, N, S, P, F are silver, gray, red, blue, yellow, gold and orange, respectively.

$$U_{\text{tot}} = \sum_{i=1}^N \sum_{j>i}^N \left\{ 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \frac{q_i q_j}{r_{ij}} \right\} + \sum_{\text{bonds}} k_{ij} (r_{ij} - r_{eq})^2 + \sum_{\text{angles}} k_{ijk} (\theta_{ij} - \theta_{eq})^2 + \sum_{\text{dihedrals}} k_{ijkl} (1 + \cos(n\phi - \delta)), \quad (1)$$

where q_i and q_j are the partial charges of the particles and r_{ij} is the bond length between adjacent atoms i and j . θ_{ijk} is the angle between the bonded atoms i , j and k , while ϕ and δ are the dihedral angles and the equilibrium state respectively, and k_{ijkl} indicates the torsion constant. The nonbonded interactions were modeled by Lennard–Jones potentials, generated through the Lorentz–Berthelot combination rule.⁵⁷ The van der Waals and Coulombic interactions between atoms separated by three bonds were scaled by factors of $f = 0.5$ and $f = 0.833$, which are the GAFF default values.⁵¹ All parameters employed in Equation 1 were obtained from previous studies,^{39,58} which have yielded excellent

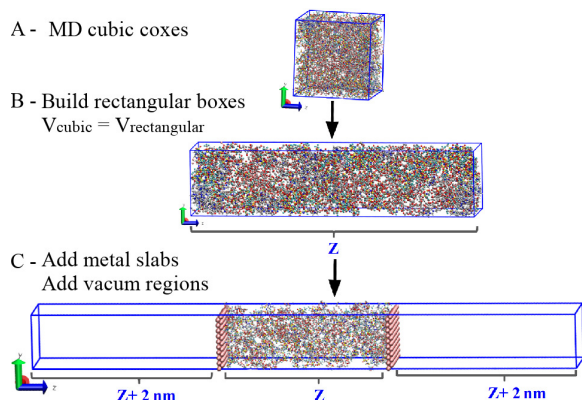


Figure 2: Schematic view of the boxes building approach. The use of $Z + 2.0 \text{ nm}$ in the vacuum region ensure total boxes Z -axis three times larger than distance between the two metal surfaces.

descriptions of the thermodynamical and structural properties of ILs-based systems^{56,58} and polymer electrolytes.^{22,39}

The atomistic models employed for the ions are shown in Figure 1, namely, 1-ethyl-3-methylimidazolium ($[\text{C}_2\text{mim}]^+$), (methyloxymethyl)triethylphosphonium ($[\text{P222mom}]^+$), 1,2,3-triazolide ($[\text{123Triaz}]^-$), poly(ethylene oxide) (PEO) chains and bis(trifluorsulphonyl)imide ($[\text{TFSI}]^-$). For PEO, an united-atom model has been employed with the hydrogen atoms being implicit accounted for in the carbon atom model.³⁵ The PEO chains consisted of 25 EO units each, yielding a molecular weight of 1174 and reasonable structural and dynamical results with low computational costs.^{22,34,35,39} For the Li-metal, we employed a slab with two Li(100) layers with body-centered-cubic symmetry with 200 atoms, which were modeled using the Lennard-Jones potential force field developed in previous work.³⁹

2.2 Box building approach

The compositions are shown in Table 1 along with the number of ions and PEO for each system. Different numbers of ions and PEO chains were used to keep similar box size along the Z -direction for the MD simulations. However, all the systems have the same ratios for Li^+ , IL and PEO, i.e. 1:1 = lithium salt:IL pair

Table 1: Number of ions and PEO chains simulated for each system, namely, $[\text{C}_2\text{mim}][\text{123Triaz}]/\text{PEO}/\text{Li123Triaz}$, $[\text{C}_2\text{mim}][\text{TFSI}]/\text{PEO}/\text{LiTFSI}$, and $[\text{P222mom}][\text{TFSI}]/\text{PEO}/\text{LiTFSI}$, which are indicated by 1, 2, 3, respectively.

System	Cation	Anion	PEO	Li^+
1	150	300	48	150
2	100	200	32	100
3	100	200	32	100

and 1:8 = $\text{Li}^+:\text{O}_{\text{ether}}$, that were chosen based on previous works.^{22,34,35,39} The simulation boxes were generated using the PACKMOL software.⁵⁹

To ensure that the lithium metal interface simulations had appropriate densities, MD simulations using cubic boxes, Figure 2A, were first carried out for all the systems and compositions given in Table 1. Once the systems reached the equilibrium, rectangular boxes, Figure 2B, with fixed volume were built using the simulated densities from the previous stage. Thereafter, the Li-metal slabs and vacuum regions were added to the box sides, i.e., interface boxes were created, Figure 2C. The X - and Y -axis boxes dimensions were locked at 3.5093 nm, while the Z -axis lengths were calculated based on the simulated densities.

Periodic boundary conditions (PBC) were used in all dimensions, and vacuum regions were hence added beyond the Li-metal slabs to ensure that no interactions occurred between the boxes and their mirror images. The vacuum regions were larger than the Z -axis to ensure that the boxes were extended to more than 3 times the initial size. Figure 2 shows the schematic view of the MD box building approach. Additional technical details are reported in the electronic supporting information, section S2.

2.3 Simulation details

The interface boxes were simulated using a 5 steps approach: (i) The initial structures were relaxed using the steepest descent algorithm; (ii) NVT stochastic simulation at 900 K over 500 ps; (iii) additional energy minimization step using

the conjugate gradient algorithm; (iv) once the negative potential energies were achieved, NVT simulations at 900 K were carried out for 4 ns, in which the temperature was decreased until 403.15 K. (v) Finally, the production was started and 230 ns of NVT simulation at 403.15 K were performed, of which the last 180 ns were used for the post-processing analysis. The atomic positions and energies were saved every 300 steps.

Temperature was maintained using a velocity-rescale thermostat⁶⁰ in the annealing simulation, while the Nose-Hoover thermostat⁶¹ was used in equilibration and production simulations with 0.1 ps coupling times. The motion equations were integrated using the leap-frog algorithm with a time step of 1 fs. The Smooth Particle Mesh Ewald method (SPME)⁶² was used to handle the Coulombic interactions and the 3dc Ewald correction⁶³ was applied to produce a pseudo-2D summation along the Z -axes.

To improve the statistic and to ensure the ergodicity of the systems, three independent simulations were carried out for each electrolyte. The initial boxes were built in three different configurations; the random box, in which all ions and polymer chains were placed randomly into the boxes, the centered and the interface boxes, in which the Li^+ ions were added to the center and the interface region of the boxes, respectively. Figure S1 shows the three starting boxes configurations used in the $[\text{C}_2\text{mim}][123\text{Triaz}]/\text{PEO}/\text{Li}123\text{Triaz}$ simulations.

3 Results and discussion

To guide the readers on the discussions, the systems are named as shown in Table 1. The structure of the systems was characterized in terms of atomic density distributions and the radial distribution functions $g(r)$ for the Li^+ environment. The PEO properties were analyzed in relation to the head-to-tail distances and radii of gyration, while the self-diffusion coefficients were calculated to get information about the ionic diffusivity.

3.1 Interface structure

To characterize the distributions of species and identify the different regions in the simulation boxes, numerical density distributions were calculated for all species relative to the center of the MD box along the Z -axis. The boxes were divided into 300 bins and the number of atoms were calculated in relation to each bin. As stated above, three independent simulations were carried out for each system, thus, the density distributions are average results, Figure 3. The independent trajectory density distributions are shown in Figures S2, S3, and S4.

3.1.1 General trends

All the ternary polymer-based electrolytes display higher densities near the Li-metal surface, i.e., there are well defined and intense peaks at about 1.0 nm from the metal surface. The densities at the interface are twice as high as in the bulk, while small oscillatory peaks are seen going from the interface to the bulk region, indicating long range interactions between the particles close to the metal surface.

The density distributions in Figure 3 show similar particles aggregation as reported by previous studies for ionic liquids interface studies.^{64–66} However, in this work we are investigating a ternary system composed by an IL, a polymer and a lithium salt. Hence, the addition of polymer and Li^+ salt does not significantly change or influence the IL ions layering capability at the surface.^{39,42–44,64}

In the three systems, the inner layers in the density distributions, i.e., the first peaks, are mainly established by the IL ions and the PEO chains. The cations and anions are quite close to the surface due to their smaller sizes and the better self-packing capability as compared to the PEO chains. While the first IL ions and the PEO peaks are located at ≈ 0.50 nm from the Li-metal surface, the Li^+ ions only appear further away in a "secondary" distributions located at about 0.50 - 0.60 nm from the surface.

The secondary Li^+ distribution seen in Figure 3 constitutes the joint result of several effects. First, there is a clear preference of the metal surface for interactions with

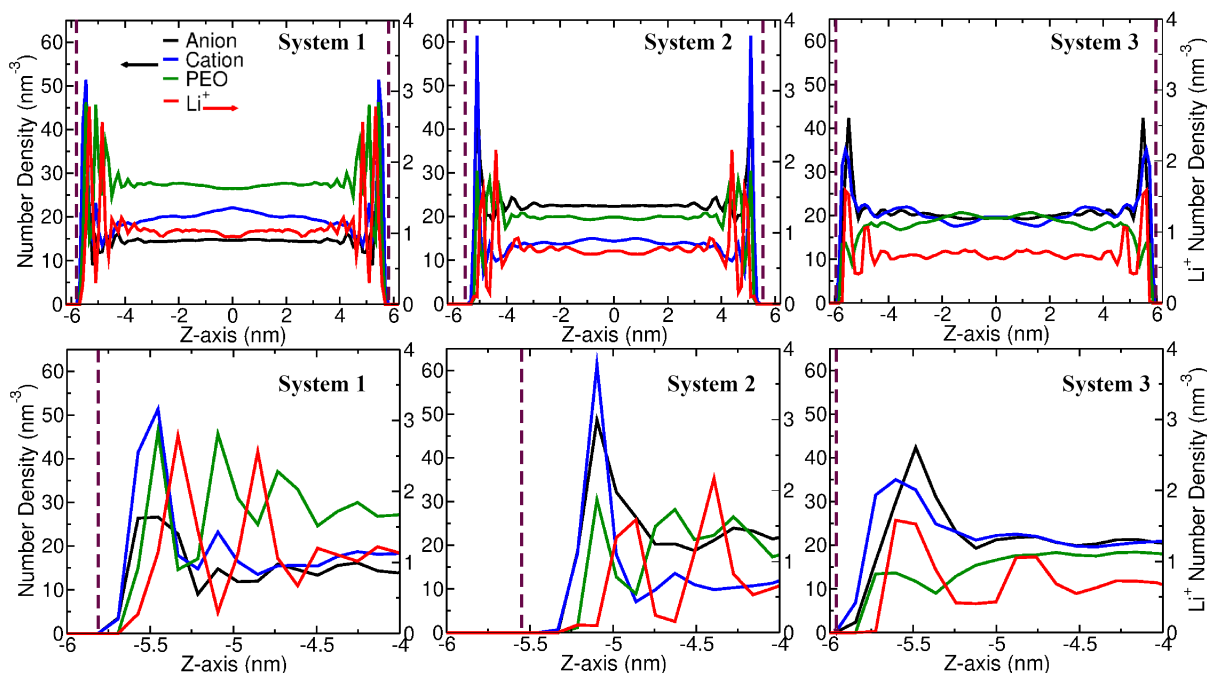


Figure 3: Top panel: number density for the systems **1**, **2** and **3** in relation to the Z -axis. Bottom panel: schematic zoom view of the interface region in the numerical density distribution for the systems **1**, **2** and **3**. The vertical dashed lines denotes the metal slab positions.

more electron-rich compounds, i.e., the IL ions and PEO chains.^{39,44} Second, besides the interactions with the surface atoms, the high concentration of IL ions at the interface leads to strong cation-anion interactions, and thus, the Li^+ can only interact with the outermost species of this inner IL layer.^{44,67} A similar effect has been seen for the lithium solvate ionic liquid $[\text{Li}(\text{G4})][\text{TFSI}]$ confined between two graphene layers investigated by Coles *et al.*⁴⁴ While the $[\text{TFSI}]^-$ anions and the G4 molecules occupy the closest space to the graphene layers, the Li^+ ions tend to appear at secondary positions, even in the negatively charged layers. Hence, the Li^+ ions rather show preferential interactions with the electrolyte species close to the surface, than to the surface atoms themselves.

3.1.2 Local trends and the effect of IL components

The bottom row of Figure 3 shows a zoom-in view of the interfacial region of the density distribution for all the investigated electrolytes. For the $[\text{C}_2\text{mim}]^+$ based systems, the inner layers are composed of sharper and more intense peaks than in the $[\text{P222mom}]^+$ based systems.

The smaller volume and the flat shape of the $[\text{C}_2\text{mim}]^+$ ⁶⁸ result in a better packing at the surface due to the orientation of the cation close to the metal surface, while the large volume and the "spherical" shape of the $[\text{P222mom}]^+$ result in a broader and more "disordered" distribution at the surface.⁶⁹

Figure 4 show the angular distribution functions (*adf*) for the angle between the normal vector of the lithium-metal surface (Z -axis) and the normal vector of the imidazolium ring planes; the vectors and θ angle are shown in the chemical structure insets in Figure 4. The *adf* shows higher populations of $[\text{C}_2\text{mim}]^+$ cations in parallel orientations close to the metal surface (i.e. large probabilities of θ at 0 being 180°) in order to maximize the interactions between the cations atoms and the Li atoms from the metal surface. In the bulk, on the other hand, this orientation is lost and a broad range $45\text{-}135^\circ$ for θ is seen. Similar behavior was observed by Ferreira and coworkers for $[\text{C}_4\text{mim}]^+$ -based ILs at gold surfaces in two different ILs, $[\text{C}_4\text{mim}][\text{TFSI}]$ and $[\text{C}_4\text{mim}][\text{PF}_6]$.⁴² In both ILs, the closest cations to the gold surface were distributed in parallel

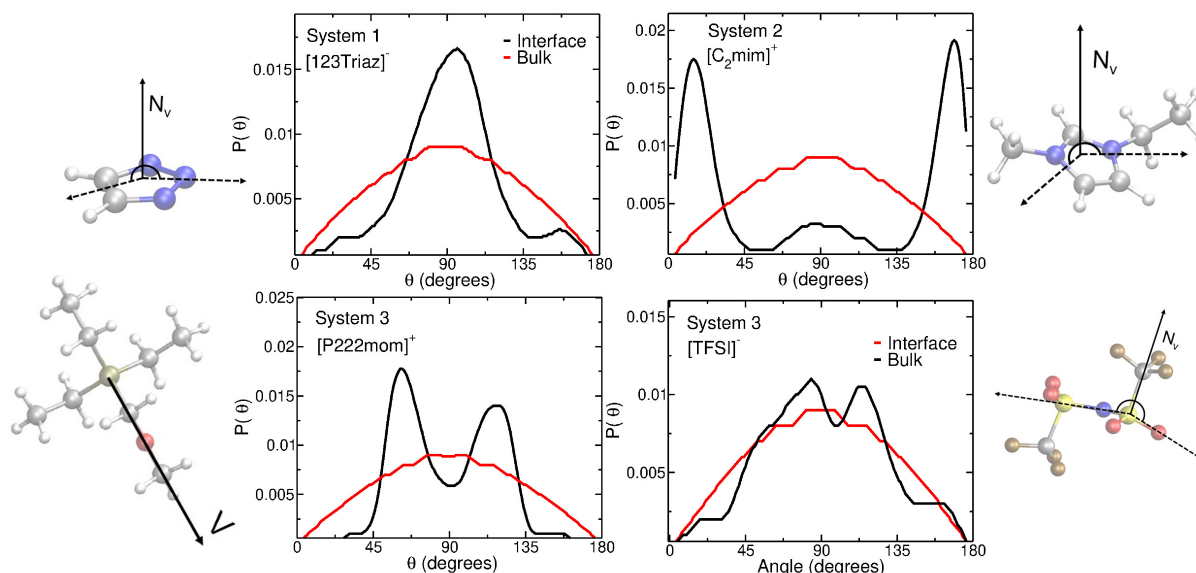


Figure 4: Angle distribution function (*adf*) for the θ angle between the normal vector from the Li-metal surface plane (S_n) and the vector defined in ions structure. $[C_2mim]^+$ in system **1** (A), $[P222mom]^+$ in system **3** (B), $[123Triaz]^-$ in system **1** (C) and $[TFSI]^-$ in system **3** (D), at the interface and bulk regions. The N_v and V vectors defined for the ions are shown in the chemical structures besides the *adf* plots, while the θ angle and the S_n vector are shown in the Li-metal surface structure (E).

stackings between the imidazolium ring and the metal surface, while the orientation is lost at longer distances and the cations instead adopt a "random" stacking.

For the angle between the normal vector of the Li-metal and the vector defined by the junction of the P and the O atoms in the $[P222mom]^+$ cation, two θ populations are seen at 60 and 120°. This is evidences of a preferential interactions between the ether oxygen from the cation and the Li atoms in the surface. However, even with the well defined $[P222mom]^+$ orientation at the surface, the cations has a broader and less structured aggregation due to the poor cation packing at the surface and their large volume. Figures S7 and S8 shown the MD snapshots of the systems **2** and **3** at the lithium-metal surface, in which a "thinner" $[C_2mim]^+$ layer can be seen in relation to $[P222mom]^+$. The cationic packing is also reflected in the species numbers in the interfacial region, for system **2**, 23.77 % of the cations are located at the interfacial region, while for **3** (the $[P222mom]^+$ based systems) this number is smaller, around 19.6 %.

Looking at Figure 3, it is seen that the differences between $[C_2mim]^+$ and $[P222mom]^+$

also affect the distribution of other particles in the systems. In system **3**, the large volume of the $[P222mom]^+$ cation leads to a broader and smaller PEO density distribution in the interface region. The replacement of the $[C_2mim]^+$ cation by $[P222mom]^+$ results in a decrease of 5.0 % in the number density of the PEO chains in the interface region due to the steric effects of the bulky $[P222mom]^+$ cation.

In the Li^+ distribution, a slight difference in the first density distribution peak position can be seen for the $[C_2mim]^+$ and the $[P222mom]^+$ based systems. Since the density distribution of PEO chains in the $[P222mom]^+$ based systems are smaller in the interface region, more Li^+ ions can come closer to the metal surface while interacting with the $[TFSI]^-$ anions and occupy the space left by PEO. The fraction of Li^+ is $\approx 4.0\%$ larger in system **3** than in system **2**. The lithium-ion population being closer to the Li-metal in system **3** is also due to the charge distribution in the $[P222mom]^+$ cation. The long alkyl chains and the ether group in the IL cation lead to a slightly delocalized negative charge around the positive P atom,

and consequently a weaker repulsion between the cation and the Li^+ .^{58,70}

Although the replacement of the IL cation in systems **2** and **3** strongly influences the PEO and Li^+ distributions, no significant effect is seen for the $[\text{TFSI}]^-$ distribution. This lack of change can be understood as a result of the strong anion-surface interactions. The replacement of the $[\text{123Triaz}]^-$ anion by the $[\text{TFSI}]^-$ in system **2** leads to some minor changes in the density distributions at the interface. The notable differences between systems **1** and **2** are primarily the broader peaks of the IL ions in system **1** due to the packing of ions in the interface region. While the $[\text{C}_2\text{mim}]^+$ tends to adopt a parallel stacking behavior at the metal surface plane, the triazolium anions are distributed in perpendicular populations, displaying a 90° θ angle between the Li surface plane and the triazolium ring plane, see Figure 4. The packing of the $[\text{123Triaz}]^-$ anions and the $[\text{C}_2\text{mim}]^+$ cations leads to a closer distribution of the electrolyte species in system **1**, and to a larger number of Li^+ ions at the interface in relation to system **2**.

Different than the other IL ions, the $[\text{TFSI}]^-$ anions do not show well defined θ angle distribution close to the surface, but only small preferences for θ at 80 and 110 and a subtle shoulder at 60. The lack of well defined peaks in the *adf* are due to the flexibility of the anion, and the multiple interactions with the Li atoms and Li^+ at same time. It is well known that the $[\text{TFSI}]^-$ can adopt different dihedral angles to optimize the intermolecular interactions with coordinating species and decrease the excluded volume around a solute. Thereby, the scattered $[\text{TFSI}]^-$ orientations at the surface constitute the combinations of several anionic conformations in the region.

3.2 Li^+ environment

The lithium-ion environment at the bulk and the interface regions were analyzed in terms of the partial radial distributions functions $g(r)$ for the interactions between the Li^+ and the most negative atoms in the (anion) and (PEO)

chains, i.e., N1, O, OE for $[\text{123Triaz}]^-$, $[\text{TFSI}]^-$ and PEO, respectively, Figure 1.

As stated above, the first electrolyte distributions extend until ca. 1.0 nm from the metal surface. Thereby, all the electrolyte particles that are within this cutoff are considered being in the interfacial region. Consequently, the electrolyte species outside of the cutoff are in the bulk region.^{42,43,64}

When studying the $\text{Li}^+ - \text{N1}([\text{123Triaz}]^-)$, $g(r)$, Figure 5, it can be seen that the peaks from the interface and the bulk regions are localized at 0.23 nm, which suggests a similar Li^+ environment in both regions. However, the higher $g(r)$ intensities in the interface region indicate a much more well-defined structure than in the bulk, mainly due to the organization of ions close to the metal surface. The $\text{N1}([\text{123Triaz}]^-)$ coordination numbers (CN) also display a significant difference when the Li^+ is close to the metal surface and in the bulk region. At the interface, the CN is almost twice the value as in the bulk, which agrees well with the $g(r)$ intensities behavior. The large density of $[\text{123Triaz}]^-$ anions at the interface and the more correlated $\text{Li}^+ - \text{N1}([\text{123Triaz}]^-)$ interaction leads to a greater anion population around the Li^+ ion.

The interface imposes a similar effect on the $\text{Li}^+ - \text{O}(\text{PEO})$, $g(r)$, as seen in the $\text{Li}^+ - \text{N1}([\text{123Triaz}]^-)$ $g(r)$, but the coordination numbers display the opposite trend. When going from the bulk to the interface region, there is a decrease in the PEO CN to Li^+ ; i.e., the PEO chains that coordinate to the Li^+ in the bulk are replaced by $[\text{123Triaz}]^-$ anions at the interface. Similar behavior was seen in previous work concerning the interface of the lithium metal and the $\text{Li}[\text{TFSI}]/\text{PEO}$ polymer electrolyte.³⁹ In the interface region, there is a depletion of the PEO chains around the Li^+ , and consequently an increase in the number of $[\text{TFSI}]^-$ anions in the lithium-ion solvation shell. Thereby, there is a larger preference for coordination to the anion by Li^+ in the interface than in the bulk.

Similar to the $\text{Li}^+ - \text{N1}([\text{123Triaz}]^-)$, $g(r)$, the $g(r)$ s for the $\text{Li}^+ - \text{O}[\text{TFSI}]^-$ interactions in systems **2** and **3**, also display sharp and intense

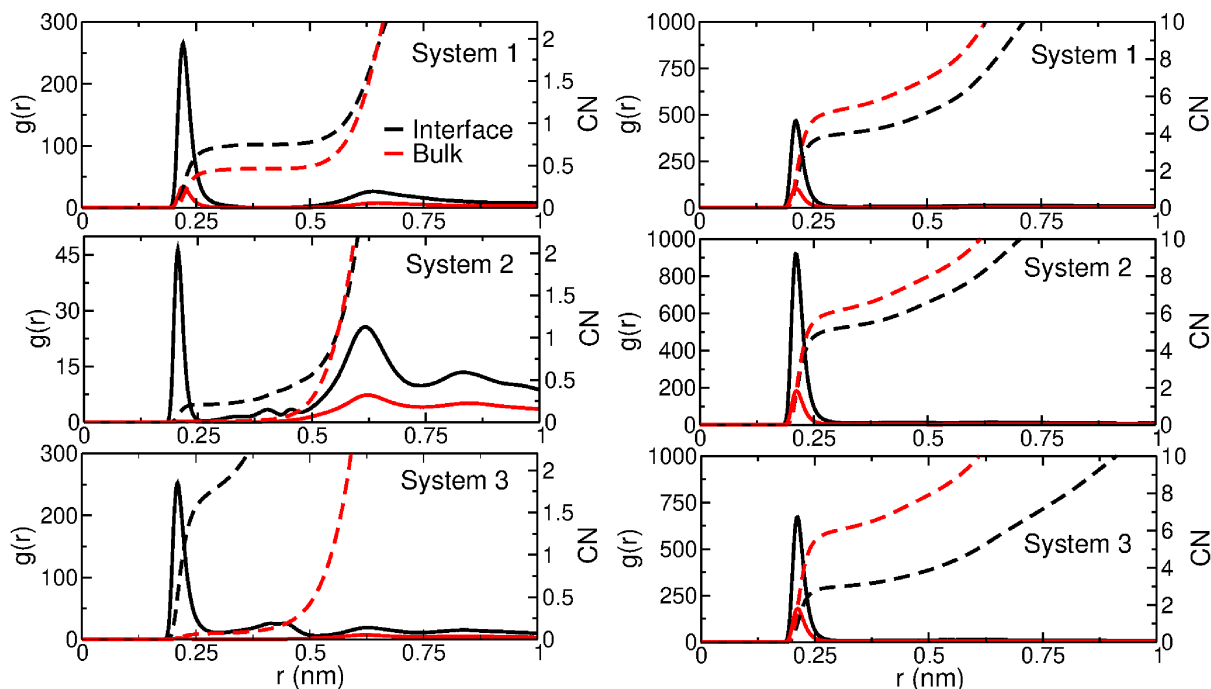


Figure 5: Partial radial distribution function ($g(r)$) and the coordination numbers (CN) at the bulk and interface regions for the interactions; left side shows $\text{Li}^+ - \text{N1}[123\text{Triaz}]^-$ in system **1** and $\text{Li}^+ - \text{O}[\text{TFSI}]^-$ in systems **2** and **3**. Right side is the partial radial distribution function ($g(r)$) for the interaction $\text{Li}^+ - \text{O}(\text{PEO})$ in systems **1**, **2** and **3**. The dashed lines are the coordination numbers obtained by the $g(r)$ integration as described in the supporting information.

peaks at 0.23 nm,²² which have higher intensities at the interface than in the bulk. However, for the $[\text{TFSI}]^-$ based systems, the interface effect is much more pronounced: in the bulk there is no direct $\text{Li}^+ - \text{O}[\text{TFSI}]^-$ interactions, while at the interface there are ≈ 2 $\text{O}[\text{TFSI}]^-$ around each Li^+ . The difference between the $\text{Li}^+ - [123\text{Triaz}]^-$ and $\text{Li}^+ - [\text{TFSI}]^-$ interactions in the bulk can be explained in terms of the chemical structure of the anions. Because of its smaller size, the triazolidine anion can occupy positions closer to the PEO chains, and thereby form direct and stronger interactions, while the larger volume of the $[\text{TFSI}]^-$ anion renders it more difficult to enter the Li^+ solvation shell in the bulk.

The partial $g(r)$ s for the $\text{Li}^+ - \text{O}(\text{PEO})$ interactions in the interface and bulk regions, shown in Figure 5, present similar structures in all systems with sharp and intense peaks at 0.23 nm. In the bulk, the Li^+ ions are coordinated by approximately 6 oxygens from the PEO chains. These values agree with previous works^{22,22,39} and are typical indicative

of the $\text{Li}^+ - \text{PEO}$ "crown" structure, Figure S11. This structure is known to favor Li^+ transport due to the segmental motion of the polymer chains which provide new sites for Li^+ to migrate into.

3.3 Polymer properties

The polymer head-to-tail distances were calculated for the PEO chains to observe changes in the polymer conformations at the interface and the bulk, as well as the influence of the different ILs on the polymer structure. The head-to-tail distances were computed by,

$$\langle r^2 \rangle^{\frac{1}{2}} = \langle |\vec{r}_n - \vec{r}_0| \rangle^2, \quad (2)$$

where $\vec{r}_n$ and $\vec{r}_0$ denote the positions of the last and the first atoms in the polymer chain.

Table 2 shows the head-to-tail distances for PEO in the interface and bulk regions. For all investigated systems, the polymer chains have a larger head-to-tail distance at the interface than in the bulk due to the

Table 2: Head-to-tail distance, $\langle r^2 \rangle^{\frac{1}{2}}$ (nm) and radius of gyration $\langle s^2 \rangle^{\frac{1}{2}}$ (nm) for the PEO chains at the interface and bulk region. Self-diffusion coefficients for the ions at interface D_{ion}^{Inter} ($10 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$) and bulk region, D_{ion}^{Bulk} ($10 \times 10^{-7} \text{ cm}^2 \text{ s}^{-1}$).

System	$\langle r^2 \rangle_{Int}^{\frac{1}{2}}$	$\langle r^2 \rangle_{Bulk}^{\frac{1}{2}}$	$\langle s^2 \rangle_{Int}^{\frac{1}{2}}$	$\langle s^2 \rangle_{Bulk}^{\frac{1}{2}}$	D_{Int}^+	D_{Int}^-	$D_{Int}^{Li^+}$	D_{Bulk}^+	D_{Bulk}^-	$D_{Bulk}^{Li^+}$
1	2.35	2.08	0.86	0.80	4.27	1.42	1.70	2.25	1.88	1.54
2	2.30	2.20	0.84	0.77	2.98	7.58	0.34	3.21	1.81	1.46
3	2.25	2.14	0.85	0.78	0.27	2.25	0.12	2.24	2.16	6.68

decrease in the $\text{Li}^+ - \text{O}(\text{PEO})$ interactions, and the more well-defined electrolyte structure in the interfacial region. With the Li^+ being coordinated in a "crown" structure in which the polymer chains are contracted in a spiral conformation, the decrease of these interactions leads to a more "extended" or "flat" PEO chain conformation, which is reflected in the larger head-to-tail distances in the interface region.

The $[\text{123Triaz}]^-$ based system has a more pronounced effect on the PEO conformations among all the investigated systems. When going from the bulk to the interface region, there is an increase of 22 % in the head-to-tail distances, while in the other systems the increase is only about 5 %. Again, the flat shape of the $[\text{123Triaz}]^-$ leads to a perpendicular stacking of the anions to the metal surface. Then, the structure of the electrolyte at the interface results in a "flatter" PEO than at the bulk (or in the other systems), and $[\text{123Triaz}]^-$ therefore has a better plasticizing effect than $[\text{TFSI}]^-$. The replacement of $[\text{C}_2\text{mim}]^+$ cations by $[\text{P222mom}]^+$ does not seem to influence the polymer structure, which is expected considering their low coordination strength.

The structure of the PEO chains was also analyzed in terms of the radius of gyration, which can be calculated from the equation below,

$$\langle s^2 \rangle^{\frac{1}{2}} = \left\langle \frac{1}{N} \sum_i |\vec{r}_i - \vec{r}_{cm}|^2 \right\rangle, \quad (3)$$

in which $\vec{r}_{cm}$ is the center of mass of a given PEO chain and $\vec{r}_i$ is the first atom in the same polymer chain. As for the head-to-tail distances, the radius of gyration in Table 2 presents larger values at the interface, mainly due to the

"flatter" polymer chains in this region. However no significant effect from the replacement of the ionic liquids ions in the systems is seen. Since the radii of gyration is the joint result from several effects, e.g., ions-PEO interactions, heat-to-tail distances, ions dynamics, etc., it is difficult to distinguish any clear effect of the ILs on the polymer properties.

3.4 Ions diffusion

The performance of a lithium-ion electrolyte is correlated to the Li^+ diffusivity, which is dependent on several factors: the breaking/formation of the Li^+ -solvent interactions, the nanostructure of the electrolyte and the interactions between the lithium-ion and the electrodes materials. Here, the transport of Li^+ , as well as other species in the systems, were analyzed by the self-diffusion coefficient, D_i , for the center of mass of the ions using the Einstein relation:⁷¹

$$D = \frac{1}{6} \lim_{x \rightarrow \infty} \frac{d}{dt} \left\langle \sum_{i=1}^N [|\vec{r}_i(t) - \vec{r}_i(0)|]^2 \right\rangle, \quad (4)$$

in which the position of the particle's center of mass is denoted by $\vec{r}_i$ at a given time t . The self-diffusion coefficients of the PEO chains were not taken into account due to the low mobility of the polymer. Due to that, the charge-neutral polymer does not contribute to the overall conductivity of the electrolyte.²⁵

Table 2 shows the self-diffusion coefficients for all the ions in the bulk and interface regions, respectively. In the bulk, the self-diffusion coefficients of systems **1** and **2** follow the typical behavior of an IL electrolyte system; i.e., cation > anion > Li^+ . As explained in several

studies,^{58,72–74} this trend is mainly induced by the strong Li^+ –anion interactions. When added to an IL, the Li^+ ions interact strongly with several anions, leading in the formation of several $[\text{Li}_x^+\text{anion}_x]$ clusters, which slow down the ionic diffusion. On the other hand, the corresponding decrease of the IL cation-cation interaction leads to that more "free" IL cations, and hence to the largest cation self-diffusion coefficient. However, system **3** follows the trend: $\text{Li}^+ > \text{cation} > \text{anion}$.

In the interfacial region, there is a decrease in the self-diffusion coefficients for all particles due to the strong interactions between the electrolyte ions and the metal surface, as well as because of the larger number of the strong Li^+ –anions interactions in the region. Also, the typical trend of D^{ions} in the IL-based electrolytes is lost and no equivalent trend can be detected between the systems.

The performance of a lithium electrolyte is generally correlated to the Li^+ diffusion in the system. Thereby, by observing the lithium-ion self-diffusion coefficients in Table 2, some interesting features can be highlighted. In the bulk, the $[\text{P222mom}]^+$ based system presents the highest D^{Li^+} , while the other two systems display similar values. As shown before, the addition of PEO into IL-based electrolytes results in a general increase of D^{Li^+} due to the weakening of the Li^+ –anion interactions and the contribution of polymer dynamics to the Li^+ diffusion mechanism. Hence, the higher $D_{\text{Bulk}}^{\text{Li}^+}$ in system **3** can be understood as a result of the larger PEO populations in the bulk of this system as compared to the other two.

At the interface, the $[\text{123Triaz}]^-$ based system presents the highest D^{Li^+} , which is correlated to the local structure of the ions in the interface. The flat $[\text{C}_2\text{mim}]^+$ and $[\text{123Triaz}]^-$ adopt specific stacking in the interface region, and therefore, the well-structured system leads to a larger free volume and formation of "diffusion channels" into the electrolyte structure.⁵⁸ Consequently, a comparatively low diffusion barrier and large self-diffusion coefficients exist in the interfacial region in this system. Therefore, the presence of flat ions and consequently the formation of specific liquid

structure at the interface region is a simple approach to induce larger Li^+ distributions and diffusivities at the interface region.

3.5 Insights into the interface effects

The sum of the results shown above helps generating a simplified picture of the interfacial structure of these IL/polymer-based ternary electrolytes at the Li-metal surface. First, the density distribution functions show a significant aggregation of the salt and IL electrolyte species at the metal surface in all systems, in which the primary aggregation, i.e., the inner distributions – are mainly composed by the more electron-rich compounds.^{39,44} However, the choice of ionic liquid ions has a direct effect on the interfacial composition. System comprising smaller IL ions generate a comparatively high PEO density distribution close to the metal surface, while systems with larger ions, like the $[\text{P222mom}]^+$ cations, result in lower polymer density distributions and a Li^+ distribution closer to the metal surface.

At the interface, the IL ions tend to adopt specific orientations to the metal surface plane due to the strong interactions between them. For example, the $[\text{C}_2\text{mim}]^+$ cations adopt parallel stackings, while the $[\text{123Triaz}]^-$ anions are distributed in perpendicular populations. In the bulk, the Li^+ environment is mainly dominated by the PEO chains with strong and close Li^+ –O(PEO) interactions. Close to the metal surface, however, there is a decrease in the number of such interactions and a corresponding increase of the Li^+ –anions coordination due to the large anion distribution in this region.

Taking the density distributions, the Li^+ environment, and the self-diffusion coefficients into account, a pattern appears. At the interface, the Li^+ diffusion mechanism is mainly dependent on the IL ionic structure due to the large ionic density distributions and the strong Li^+ –(anions) interactions. In contrast, in the bulk of the ternary electrolyte, the PEO chains exert a larger effect on the Li^+ dynamics, mainly due to the larger PEO population in this region. This agrees well with the previous bulk PEO

electrolyte studies.^{22,25,34} Therefore, the choice of the IL and its content can be used in the optimization of the interfacial nanostructure, which in turn can control Li⁺ transport both in the bulk and in the interfacial region.

4 Conclusions

In the present work, classical MD simulations were carried out for three IL ternary based electrolytes confined between two lithium-metal surfaces: [C₂mim][123Triaz]/PEO/Li123Triaz, [C₂mim][TFSI]/PEO/LiTFSI and [P222mom][TFSI]/PEO/LiTFSI. The number density distributions were calculated to characterize the distribution of particles in the box and to identify the electrolyte interfacial region close to the metal surface. The dynamics of the ions were investigated in terms of the self-diffusion coefficients, while the PEO properties were characterized by the radius of gyration and head-to-tail distances.

All systems shown the classical IL-based electrolyte behavior, with a significant aggregation of ions at the interface. In the interface region, the interactions between the Li surface atoms and the electrolyte species result in a highly specific local structure. The analysis thereby shows the possibility to tune the electrolytes performance and the interfacial properties through the choice of the IL, taking ions size, shape and volume into account. Among all the studied systems in the present work, the [C₂mim][123Triaz]/PEO/Li123Triaz shows the best potential as IL-based electrolyte candidate for lithium-metal batteries, based on the Li⁺ populations at the interface and their self-diffusion coefficients in the region.

Although we have performed classical MD simulations, i.e., no description of reactions or electrolyte decomposition, the results above display some important indications which could be related to SEI formation in these systems, mainly due to the density distributions and Li⁺ properties at the interface. Thereby, future studies which employ different computational techniques, such as reactive molecular dynamics and quantum calculations, are encouraged. We

also would like to highlight that due to the nature of the Li-metal model used in this work, i.e. simple Lennard–Jones models, but, the conclusions obtained here, may be extended to different interface metal materials. Thereby, this work is also contributing to the general IL-based electrolyte-metal interface studies.

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Supporting Information Available

MD cubic boxes simulation approach; interface box building; systems **1**, **2** and **3** numerical density distributions: random, center, interface and average; interface region composition; interface snapshots systems **1**, **2** and **3**; PEO/Li⁺ solvation structure snapshot. The GROMACS force field input files are attached to the supporting information.

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Graphical TOC Entry

